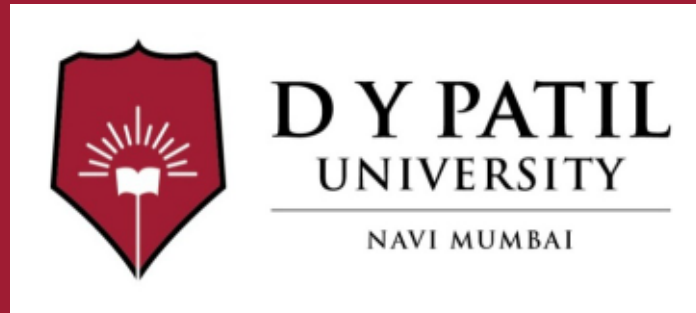




Statistical Techniques

Unit No 4 : Regression Analysis

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Syllabus

Unit IV: Regression Analysis

- Concept of Regression
- Importance of Regression Analysis in Computer Application.
- Calculation of Regression Coefficient
- Calculation of Regression Equations:
Regression equation of Y on X and Regression Equation of X on Y,



Concept of Regression



Meaning of Regression

Regression is a statistical technique used to study the relationship between variables and predict one variable based on another.

What does Regression do?

It helps answer questions like:

How does **X** affect **Y**?

Can we **predict Y** if we know **X**?

Example:

If advertising increases, will sales increase?

Regression helps find that relationship.



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Meaning of Regression

Regression is a statistical technique used to study the relationship between variables and to predict the value of one variable based on another.

In Simple Words

“Regression = prediction using relationship”



Key Idea

- One variable is **dependent (Y)** → what we want to predict
- One or more variables are **independent (X)** → factors influencing Y



Why is Regression Important?

- Helps in prediction & forecasting
- Identifies relationships between variables
- Supports data-driven decision making



Importance of Regression Analysis in Computer Applications



Importance of Regression Analysis in Computer Applications

Regression analysis is a statistical technique used to understand the relationship between variables and to predict future outcomes. In computer applications, it plays a very important role in data-driven decision-making.

1. Prediction and Forecasting

Regression analysis helps in predicting future values based on past data.

It is used to forecast sales, demand, or user behavior in software systems.

It helps applications make intelligent predictions automatically.

It improves planning and decision-making in business applications.

Examples:

An e-commerce website predicts future product sales based on past purchase data.

A weather app predicts temperature using historical climate data.

A stock market application predicts price trends using previous stock data.



Importance of Regression Analysis in Computer Applications

2. Data Analysis and Relationship Understanding

Regression helps in identifying relationships between variables in datasets.

It shows how one variable affects another.

It helps developers and analysts understand patterns in data.

It supports better design of algorithms and systems.

Examples:

An app analyzes how study hours affect student performance.

A fitness app studies the relationship between exercise time and calories burned.

A marketing system analyzes how advertising cost affects sales.



Importance of Regression Analysis in Computer Applications

3. Machine Learning and AI Applications

Regression is widely used in machine learning models.

It is a basic technique in supervised learning.

It helps in building predictive models in AI systems.

It improves accuracy of intelligent applications.

Examples:

A recommendation system predicts user ratings for movies or products.

A chatbot predicts user intent based on input data.

A spam detection system predicts whether an email is spam or not.



4. Performance Optimization

Regression helps in improving system performance.

It identifies factors affecting system efficiency.

It helps optimize algorithms and resources.

It improves speed and accuracy of applications.

Examples:

A website analyzes how server load affects response time.

A mobile app studies battery usage based on app activity.

A cloud system predicts resource usage for better allocation.

5. Risk Analysis and Decision Making

Regression supports better decisions by analyzing risks.

It helps identify potential risks in systems.

It supports data-driven decision-making.

It reduces uncertainty in business and technical environments.

Examples:

A banking system predicts loan default risk.

A cybersecurity system predicts chances of cyber attacks.

An insurance app estimates risk based on user data.

Conclusion

Regression analysis is very important in computer applications because it helps in prediction, understanding data, improving systems, and making better decisions using data.

Difference between Correlation and Regression

Basis of Distinction	Correlation	Regression
1. What measures?	Correlation measures the degree and direction of relationship between variables.	Regression measures the nature and extent of average relationship between two or more variables in terms of original units.
2. Relative or absolute measure	It is a relative measure showing association between variables.	It is an absolute measure of relationship.
3. Independence of origin and scale	Correlation coefficient is independent of change of both origin and scale .	Regression coefficient is independent of origin but not of scale .
4. Units of measurement	Correlation coefficient is independent of units of measurement .	Regression coefficient is dependent on units of measurement .
5. Expression of relationship	Value ranges from -1 to +1 .	Expressed in equation form like: $Y = a + bX$ or $Y = a + bX + cX^2$.
6. Forecasting device	It is not a forecasting tool .	It is a forecasting tool used to predict dependent variable values.



Example 1

A software company wants to analyze whether a programmer's performance in an internal coding test is related to their project performance score.

The following data represents:

X → Coding Test Score (out of 50)

Y → Project Performance Score (out of 50)

X	Y
40	32
34	39
28	26
30	30
44	38
38	34
31	28



Questions

Q1. Find the regression coefficients:

b_{yx} (regression coefficient of Y on X)

b_{xy} (regression coefficient of X on Y)

Q.2 Calculate the regression equation of Y on X (Project Performance on Coding Score).
Calculate the regression equation of X on Y (Coding Score on Project Performance).

Q.3 Interpret the meaning of the regression coefficient:

What does it indicate about the relationship between coding ability and project performance?

Q.4 If a student scores 36 marks in the coding test, estimate their expected project performance score using the regression equation.

Q.5 Based on your analysis: Should the company rely on coding test scores to predict project performance? Justify your answer with statistical reasoning.



Calculation of Mean of X

Given values of X:

40, 34, 28, 30, 44, 38, 31

Step 1: Find sum

$$\sum X = 40 + 34 + 28 + 30 + 44 + 38 + 31 = 245$$

Step 2: Apply formula

$$\bar{X} = \frac{\sum X}{n}$$

$$\bar{X} = \frac{245}{7} = 35$$

Mean of X = 35

Calculation of Mean of Y

Given values of Y:

32, 39, 26, 30, 38, 34, 28

Step 1: Find sum

$$\sum Y = 32 + 39 + 26 + 30 + 38 + 34 + 28 = 227$$

Step 2: Apply formula

$$\bar{Y} = \frac{\sum Y}{n}$$

$$\bar{Y} = \frac{227}{7} = 32.43$$

Mean of Y = 32.43



Q1. Find the regression coefficients:
byx (regression coefficient of Y on X)
bxy (regression coefficient of X on Y)

X	Y	(x = X-35)	(y = Y -32.43)	(xy)	(x ²)	(y ²)
40	32	5	-0.43	-2.15	25	0.18
34	39	-1	6.57	-6.57	1	43.16
28	26	-7	-6.43	45.01	49	41.34
30	30	-5	-2.43	12.15	25	5.90
44	38	9	5.57	50.13	81	31.02
38	34	3	1.57	4.71	9	2.46
31	28	-4	-4.43	17.72	16	19.62



Summations:

$$\sum x = 0, \quad \sum y = 0$$

$$\sum xy = 121$$

$$\sum x^2 = 206$$

$$\sum y^2 = 143.68$$



Q1. Find the regression coefficients:
byx (regression coefficient of Y on X)
bxy (regression coefficient of X on Y)

Regression Coefficients

$$b_{yx}$$

$$b_{yx} = \frac{\sum xy}{\sum x^2} = \frac{121}{206} = 0.587$$

$$b_{xy}$$

$$b_{xy} = \frac{\sum xy}{\sum y^2} = \frac{121}{143.68} = 0.842$$



Q. 2 Calculate the regression equation of Y on X (Project Performance on Coding Score).

Regression Equation of Y on X

$$y - \bar{Y} = b_{yx}(x - \bar{X})$$

Where:

$$b_{yx} = \frac{\sum xy}{\sum x^2}$$

$$\sum xy = 121, \quad \sum x^2 = 206,$$

$$b_{yx} = \frac{121}{206} = 0.587$$

$$Y - 32.43 = 0.587(X - 35)$$

$$Y - 32.43 = 0.587X - 20.545$$

$$\mathbf{Y = 0.587X + 11.885}$$



Calculate the regression equation of X on Y (Coding Score on Project Performance).

$$b_{xy} = \frac{\sum xy}{\sum y^2}$$

$$b_{xy} = \frac{121}{143.68} = 0.842$$

$$X - \bar{X} = b_{xy}(Y - \bar{Y})$$

$$X - 35 = 0.842Y - 27.3$$

$$\mathbf{X = 0.842Y + 7.7}$$



Q.3 Interpret the meaning of the regression coefficient:

What does it indicate about the relationship between coding ability and project performance?

The regression coefficient $b_{yx}=0.587$ indicates a positive and moderate relationship between coding ability and project performance.

The regression coefficient $b_{xy}=0.842$ indicates a positive relationship between project performance and coding ability.



Q.4 If a student scores 36 marks in the coding test, estimate their expected project performance score using the regression equation.

$$X=36$$

$$Y=21.132+11.885$$

$$Y=33.017$$

$$Y \approx 33$$

When coding test score is **36**, the expected project performance score is approximately **33 marks**.



Q.5 Based on your analysis: Should the company rely on coding test scores to predict project performance? Justify your answer with statistical reasoning.

Should the company rely on coding test scores to predict project performance?

Partially yes, but not completely.

Statistical Justification

1. Positive Relationship Exists

Regression coefficient:

$$b_{yx}=0.587$$

Indicates a **positive relationship**

Higher coding scores → higher project performance

Regression gives prediction, not perfect accuracy



Example 2

company wants to analyze whether programmer experience affects the number of bugs produced in code.

X → Years of Programming Experience

Y → Number of Bugs in Code

X (Years)	Y (Bugs)
1	50
2	45
3	40
4	35
5	30
6	28
7	25

Questions

Q1. Find the regression coefficients:

b_{yx} (regression coefficient of number of bugs on experience)

b_{xy} (regression coefficient of experience on number of bugs)

Q2. Calculate the regression equations:

Find the regression equation of **Y on X** (Number of Bugs on Experience)

Find the regression equation of **X on Y** (Experience on Number of Bugs)

Q3. Interpret the meaning of the regression coefficient:

What does it indicate about the relationship between programmer experience and **number of bugs produced**?

Q4. If a programmer has 4 years of experience, estimate the expected number of bugs using the regression equation.

Q5. Based on your analysis:

Should the company rely on programmer experience to predict the number of bugs in code? Justify your answer with statistical reasoning.



Answer

Mean

$$\bar{X} = \frac{1 + 2 + 3 + 4 + 5 + 6 + 7}{7} = 4$$

$$\bar{Y} = \frac{50 + 45 + 40 + 35 + 30 + 28 + 25}{7} = 36.14$$



X	Y	$X - \bar{X}$	$Y - \bar{Y}$	$(X - \bar{X})^2$	$(Y - \bar{Y})^2$	$(X - \bar{X})(Y - \bar{Y})$
1	50	-3	13.86	9	192.10	-41.58
2	45	-2	8.86	4	78.50	-17.72
3	40	-1	3.86	1	14.90	-3.86
4	35	0	-1.14	0	1.30	0
5	30	1	-6.14	1	37.70	-6.14
6	28	2	-8.14	4	66.30	-16.28
7	25	3	↓ 11.14	9	124.10	-33.42



Summations

$$\sum (X - \bar{X})^2 = 28$$

$$\sum (Y - \bar{Y})^2 = 515$$

$$\sum (X - \bar{X})(Y - \bar{Y}) = -119$$

Regression Coefficients

b_{yx}

$$b_{yx} = \frac{-119}{28} = -4.25$$

b_{xy}

$$b_{xy} = \frac{-119}{515} = -0.231$$

Answer Q. 2

Regression Equations

► Y on X:

$$Y - 36.14 = -5(X - 4)$$

$$Y = 36.14 - 5(X - 4)$$

► X on Y:

$$X - 4 = -0.2(Y - 36.14)$$

$$X = 4 - 0.2(Y - 36.14)$$

Answer : Q3

Q3. Interpret the meaning of the regression coefficient:

What does it indicate about the relationship between programmer experience and number of bugs produced?

Answer :

The regression coefficient $b_{yx} = -4.25$ indicates a negative relationship between programmer experience and the number of bugs.

This means that for every 1-year increase in experience, the number of bugs decreases by approximately 4.25 units.

The negative sign shows an inverse relationship:

As experience increases → bugs decrease

As experience decreases → bugs increase

Negative regression coefficient indicates an inverse relationship. As programmer experience increases, the number of bugs decreases. The value shows that experience significantly reduces bugs.

Answer : Q. 4

Experience $X = 4$

Regression equation of Y on X:

$$Y = 36.14 - 4.25(X - 4)$$

Substitute $X = 4$

$$Y = 36.14 - 4.25(4 - 4)$$

$$Y = 36.14 - 4.25(0)$$

$$Y = 36.14$$

The expected number of bugs $\approx$ **36**



Answer : Q. 5

Q5. Based on your analysis:

Should the company rely on programmer experience to predict the number of bugs in code? Justify your answer with statistical reasoning.

Answer :

Yes, the company can rely on programmer experience to some extent for predicting the number of bugs.

Justification (Statistical Reasoning):

The regression coefficient $b_{yx} = -4.25$ is negative, indicating a strong inverse relationship between experience and number of bugs.

This means that as experience increases, the number of bugs consistently decreases.

Therefore, experience is a **useful predictor** of code quality.



Calculation of Regression when Correlation Coefficient is given



Regression Equations (Y on X and X on Y)

In regression, we often form **two equations**:

Regression of Y on X → used to predict **Y from X**

Regression of X on Y → used to predict **X from Y**

◆ 1. Regression Equation of Y on X

Formula:

$$Y - \bar{Y} = b_{yx}(X - \bar{X})$$

or

$$Y = a + bX$$

Where:

- $b_{yx} = r \cdot \frac{\sigma_Y}{\sigma_X}$

◆ 2. Regression Equation of X on Y

Formula:

$$X - \bar{X} = b_{xy}(Y - \bar{Y})$$

or

$$X = a + bY$$

Where:

- $b_{xy} = r \cdot \frac{\sigma_X}{\sigma_Y}$



Example 1

Given data:

Mean of $X = 10$

Mean of $Y = 20$

Standard deviation of $X = 2$

Standard deviation of $Y = 4$

Correlation (r) = 0.5

Find

1. Régression Coefficient (Y on X) and Régression Coefficient (X on Y)
2. Régression Equations (Y on X) and (X on Y)



Solution:

✓ Step 1: Regression Coefficient (Y on X)

$$b_{yx} = r \cdot \frac{\sigma_Y}{\sigma_X} = 0.5 \cdot \frac{4}{2} = 1$$

Equation:

$$Y - 20 = 1(X - 10)$$

$$Y = X + 10$$

✓ Step 2: Regression Coefficient (X on Y)

$$b_{xy} = r \cdot \frac{\sigma_X}{\sigma_Y} = 0.5 \cdot \frac{2}{4} = 0.25$$

Equation:

$$X - 10 = 0.25(Y - 20)$$

$$X = 0.25Y + 5$$



Example 2

Given:

Mean of $X = 5$, Mean of $Y = 10$

$$\sigma_X = 2, \sigma_Y = 4$$

$$r = 0.6$$

find Régression Coefficient (Y on X) and
Régression Coefficient (X on Y) and Régression
Equations

Solution:

◆ Step 1: Y on X

$$b_{yx} = r \cdot \frac{\sigma_Y}{\sigma_X} = 0.6 \cdot \frac{4}{2} = 1.2$$

$$Y - 10 = 1.2(X - 5)$$

$$Y = 1.2X + 4$$

◆ Step 2: X on Y

$$b_{xy} = 0.6 \cdot \frac{2}{4} = 0.3$$

$$X - 5 = 0.3(Y - 10)$$

$$X = 0.3Y + 2$$



Thank You



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